



✓ **Addendum** to the e-commerce site ERD, incorporating messaging and reviews, with **all the modification notes**:

1. **Users can only message someone if they've had business together** (i.e., there must be a completed transaction).
2. **Messages are versioned**:
 - Users **cannot delete messages** after they are sent.
 - Users can **edit messages**, creating **new versions**.
 - The **latest version** is displayed by default.
 - **Version history** is maintained.
3. **Users can only review a product or vendor once.**
4. **Users can only leave reviews if they've had business together.**
5. **Users retain full CRUD privileges over their reviews** (Create, Read, Update, and Delete).

ERD Description with All Modifications

1. Users

- user_id (PK) – Unique identifier
- first_name – User's first name
- last_name – User's last name
- email – User's email address
- password – Hashed password
- created_at – Account creation date
- role – Enum (customer, vendor, administrator)
- shopping_cart (JSON or serialized collection) – Array of product objects with:
 - product_id – ID of the product in the cart
 - unit_price – Price at the time added to the cart



- quantity – Quantity added
- transactions (JSON or serialized collection) – Array of transaction IDs linked to the user
- addresses (JSON or serialized collection) – Array of address objects with:
 - address_id – Unique identifier for the address
 - street – Street name and number
 - city – City name
 - state – State or province
 - postal_code – ZIP or postal code
 - country – Country name
 - isDefault – Boolean (true if the current address is the default)
- payment_methods (JSON or serialized collection) – Array of payment method objects with:
 - payment_id – Unique identifier for the payment method
 - card_number – Masked card number (e.g., **** * 1234)
 - expiry_date – Card expiration date
 - payment_type – Enum (credit_card, debit_card, paypal, etc.)
 - isDefault – Boolean (true if it is the default payment method)
- privileges (JSON object of objects) – Object containing multiple privilege objects as properties, with:
 - isGranted – Boolean (true if the privilege is granted, false if revoked)
 - description – Description of the privilege
 - granted_by – user_id of the admin or business that granted the privilege
 - granted_at – Date and time the privilege was granted



2. Transactions

- transaction_id (PK) – Unique identifier
- user_id (FK) – Links to Users (the buyer)
- vendor_id (FK) – Links to Users (the vendor)
- total_amount – Total amount paid
- payment_method_id – References the payment method used
- transaction_date – Date and time of purchase
- shipping_address_id – References the shipping address
- billing_address_id – References the billing address (defaults to the shipping address but can be different)
- products (JSON or serialized collection) – Array of product objects with:
 - product_id – ID of the product purchased
 - unit_price – Price at the time of purchase
 - quantity – Quantity purchased
- tracking_status (Stack collection of tracking objects) – Array of tracking objects representing the delivery status stack:
 - tracking_id – Unique identifier for the tracking event
 - status – Enum (purchased, shipped, in_transit, out_for_delivery, delivered, canceled, return, returned)
 - status_date – Date and time of the status update
 - latitude – Google Maps latitude coordinate
 - longitude – Google Maps longitude coordinate



✓ Constraints:

- **Messaging requires a transaction:**
 - Users can only message someone if they are completing or have a **completed transaction** together. Once the transaction is finalized and closed, the messaging channel is deactivated.
 - **Reviewing requires a transaction:**
 - Users can only review a **product or vendor once** and only if they've had business together.
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3. Messages

- message_id (PK) – Unique identifier for the message thread
- transaction_id (FK) – Links to Transactions (ensures that only users with a business relationship can message each other)
- sender_id (FK) – Links to Users table (the sender)
- receiver_id (FK) – Links to Users table (the receiver)
- default_version_id (FK) – Points to the **latest version** of the message
- sent_at – Date and time when the original message was sent
- is_read – Boolean (true if the message has been read)
- ✓ **No Deletion:**
 - Users **cannot delete messages** after they are sent.
 - Messages remain visible permanently (only latest versions).

✓ Relationships:

- **transaction_id** → Ensures only users with a **completed or ongoing transaction** can message each other.



✓ Example Message Data:

```
{
  "message_id": "m001",
  "transaction_id": "t567",
  "sender_id": "u123",
  "receiver_id": "v456",
  "content": "{latest, archive:[older versions]}",
  "default_version_id": "v003",
  "sent_at": "2025-03-25T10:15:00Z",
  "is_read": true
}
```

4. Message Versions

- version_id (PK) – Unique identifier for each version of the message
- message_id (FK) – Links to the original message
- content – Text message content for this version
- version_number – Integer indicating the version sequence
- updated_at – Date and time when this version was created
- is_latest – Boolean (true if this is the latest version)

✓ Relationships:

- **message_id → default_version_id** ensures that the latest version is displayed by default.
- All previous versions are stored in the message_versions table.



✓ **Example Message Version Data:**

```
[
  {
    "version_id": "v001",
    "message_id": "m001",
    "content": "Hi, is this product available?",
    "version_number": 1,
    "updated_at": "2025-03-25T10:15:00Z",
    "is_latest": false
  },
  {
    "version_id": "v003",
    "message_id": "m001",
    "content": "Hi, is this product available in navy blue?",
    "version_number": 3,
    "updated_at": "2025-03-25T10:30:00Z",
    "is_latest": true
  }
]
```



5. Reviews

- review_id (PK) – Unique identifier
- user_id (FK) – Links to Users table
- vendor_id (FK) – Links to Users table (for vendor reviews)
- product_id (FK) – Links to Products table (for product reviews)
- rating – Integer (1 to 5 stars)
- comment – Text feedback {latest, archive: [older versions]}
- review_date – Date and time the review was posted
-  **CRUD privileges:**
 - Users can **Create, Read, Update, and Delete** their reviews.
-  **Constraint:**
 - Only one review per **user-product** or **user-vendor** relationship.
 - Only allowed if they've had business together.

Key Features:

1. **Transaction-verified messaging and reviews**
 2. **Message versioning with no deletion**
 3. **CRUD privileges over reviews**
 4. **Data integrity with constraints**
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